



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 4 - Day 4 Report

Internship Program — Batch 2026

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1. Introduction

Experiment No. 16 focused on Machine Learning Operations, commonly known as MLOps, which is a set of practices that combines machine learning, software engineering, and DevOps principles to reliably build, deploy, monitor, and maintain ML models in production environments. MLOps addresses the challenges of moving models from research notebooks to live applications where they can serve real users at scale. This experiment introduced the fundamental concepts and tools used in MLOps, including model serialization with joblib, API creation with FastAPI, user interface development with Streamlit, and application containerization with Docker. Understanding these concepts was essential for transitioning from experimental models to production-ready AI systems.

2. Objectives

- Understand the MLOps lifecycle and its importance in production ML deployments.
 - Learn how to save and load trained machine learning models using joblib.
 - Explore how to serve models as APIs using the FastAPI web framework.
 - Understand how to build quick user interfaces with Streamlit for model demos.
 - Learn about containerization and how Docker packages applications for consistent deployment.
 - Understand the end-to-end process of making a model accessible to end users.
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3. Task Description

Experiment No. 16 involved studying the complete MLOps workflow, from trained model to live deployment. The process of saving trained models to disk using joblib was explored, enabling models to be reused without retraining. Building a web API with FastAPI was examined, understanding how machine learning models can be exposed as endpoints that other applications can call over HTTP.

The experiment also covered Streamlit for creating quick interactive interfaces, allowing non-technical users to interact with models through sliders, buttons, and forms. Finally, Docker containerization was studied to understand how applications can be packaged with all their dependencies, enabling consistent deployment across different environments.

4. Work Done

Experiment No. 16 began with understanding the MLOps lifecycle and the challenges of moving models from development to production. The concept of model serialization was explored, learning how the joblib library saves trained models to persistent storage so they can be loaded without retraining.

The FastAPI framework was studied in detail, understanding how to define API endpoints, handle incoming requests with JSON payloads, load saved models, and return predictions as

structured responses. Code examples demonstrated how to create a complete API server that listens for prediction requests and returns model outputs.

The Streamlit library was explored as a solution for rapid interface development. Examples showed how to create interactive interfaces with input widgets, buttons, and display areas that allow users to input data and see model predictions instantly.

Docker was introduced as a containerization tool that packages applications along with their dependencies, ensuring consistent behavior across development, testing, and production environments. The concept of Dockerfiles and container builds was examined to understand how applications are packaged for reliable deployment.

5. Challenges Faced

One of the main challenges in Experiment No. 16 was understanding the relationship between different deployment components and how they work together to create a complete system. Learning how an API server interacts with a saved model, how a user interface sends requests to the API, and how containers package everything together required careful study of each component and its role in the overall architecture.

Another challenge involved understanding best practices for model serialization and versioning, ensuring that saved models can be reliably loaded across different environments and versions of libraries. These concepts were better understood through studying practical examples and deployment workflows.

6. Key Learnings

Experiment No. 16 provided valuable knowledge of MLOps fundamentals and the tools used in production ML deployments. I learned how model serialization with joblib enables models to be saved and reused, decoupling the training process from the deployment process.

The FastAPI framework showed me how machine learning models can be transformed into accessible web services, enabling integration with other applications through standardized HTTP APIs. I also learned how Streamlit simplifies the creation of user interfaces, making it easy to demonstrate model capabilities to stakeholders and non-technical users.

The introduction to Docker provided insight into application containerization and how it addresses the "it works on my machine" problem by packaging applications with all their dependencies. Understanding containerization is essential for modern software development and deployment practices.

7. Conclusion

Experiment No. 16 successfully demonstrated the essential practices and tools for deploying machine learning models to production environments. By learning how to save models, create APIs, build user interfaces, and containerize applications, I gained a solid foundation in the practical aspects of operationalizing AI systems.

The experiment highlighted the importance of MLOps in bridging the gap between experimental ML research and real-world applications. The skills learned provide a strong foundation for building production-ready AI applications that can serve users reliably at scale.

Intern Signature:



Supervisor:

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Name:

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Designation: